Everything proceeds smoothly, and when Alice gets to Alpha Centauri she indeed looks at her coin. Amazingly, the instant she looks at it, she immediately knows exactly what coin Bob will see, even before he looks. Is this crazy? Have Alice and Bob succeeded in breaking relativity's most fundamental rule, which states that information cannot go faster than the speed of light?

Of course not. What would violate relativity would be for Alice's observation *to instantly tell Bob* what to expect. Alice may know what coin Bob will see but she has no way to tell him—not without sending him a real message from Alpha Centauri, and that would take at least the four years required for light to make the trip.

Let's do this experiment many times, either with many Alice-Bob pairs or with the same pair spread out over time. In order to be quantitative, Charlie (he's back now, having accepted our apology) paints a "$\sigma = +1$" on each penny and a "$\sigma = -1$" on each dime. If we assume that Charlie really is random in the way he shuffles the coins, then the following facts will emerge:

- On average, both A and B will get as many pennies as dimes. Calling the values of A's observations $\sigma_A$ and B's observations $\sigma_B$, we can express this fact mathematically as

$$\begin{aligned} \langle \sigma_A \rangle &= 0 \\ \langle \sigma_B \rangle &= 0. \end{aligned} \qquad (6.1)$$

- If A and B record their observations and then get together back in Palo Alto to compare them, they will find a strong correlation.[1] For each trial, if A observed $\sigma_A = +1$, then B observed $\sigma_B = -1$, and vice versa. In other words, the product $\sigma_A \sigma_B$ always equals $-1$: